SESSION 9 : Understanding & Evaluating Gen AI outputs.

MODULE 1 : PROMPT ENG. & API INT.

Inputs → API → LLM → Returns → Output

Outputs have hallucination sometimes

Root Cause 1: Training Data Bias/Gaps

Root Cause 2: Next token objective limits (Fluency ≠ Truth)

Root Cause 3: Over Confidence

4: Lack of Grounding

5: vague prompts.

Taxonomy of A Hallucination.

1. Factual Fabrication.
2. Citation Hallucination

Model Limitations and Behaviors.

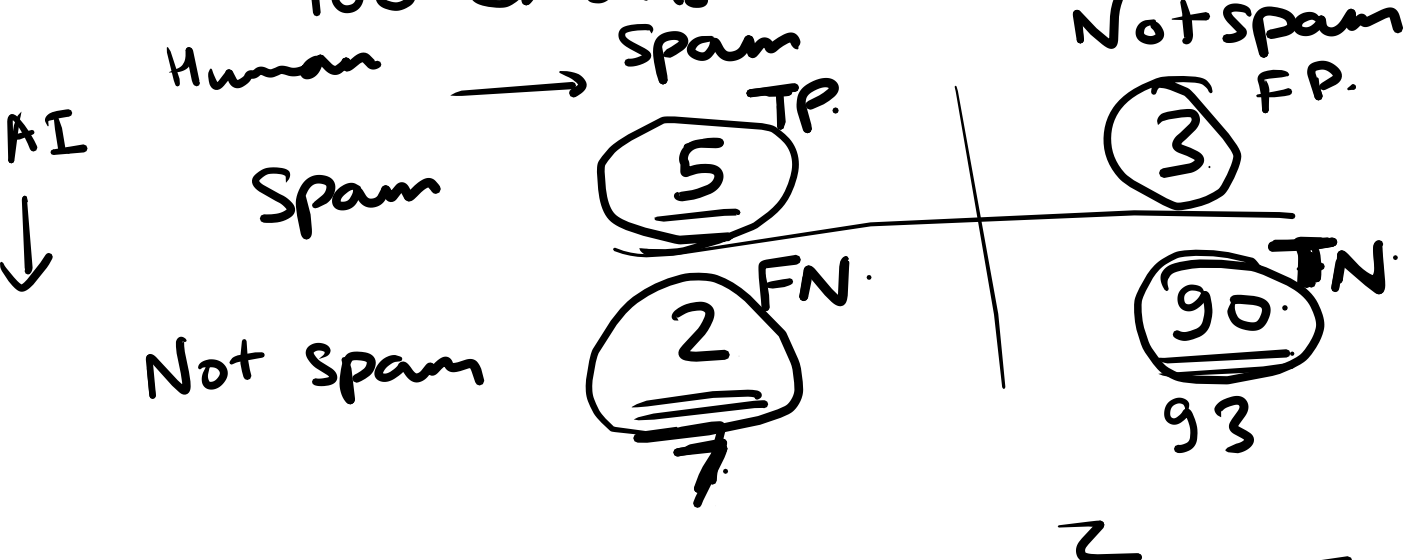
1. Structural limitation
2. Input sensitivity.
3. Behavioral patterns in Generation.
4. Numerical & Logical Fragility.
5. Quirk surfacing.

3. Evaluation Metrics

3.1 Traditional Evaluation Metrics

AI checks spam or not spam.

100 emails



$$\text{Accuracy} = \frac{90+5}{100} = 95\%$$

$$\text{Precision} = TP/(TP+FP) = 5/8 = 62.5\%$$

$$\text{Recall} = TP/(TP+FN) = 5/7 = 71\%$$

$$\text{F1 score} = \frac{2}{\left(\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}\right)} = \frac{2}{\frac{8}{5}+\frac{7}{5}} = \frac{2}{3} = 66.6\%$$

2. NLP BLEU / ROGUE / METEOR.

How AI words were used by Human also?

Is AI covering enough content from Human?

Covers Synonyms.

3 Regression

MAE → How close were our numbers

RMSE →

LLM Metric Layering Model.

1. Automated Signals
   - Structure Validity = 80%.
   - Destination Accuracy
   - Repetation Checks
   - Time Allocation.
   - Output lengths.

   Programatic checks.

2. Human judgement
   - Intent. = 90%
   - Engage = 90%
   - Novelty
   - Cultural Sensitivity.

3. Aggregation.